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GitHub Link	https://github.com/suri307/MLA0403-DEEP-LEARNING/tree/main/ASSESSMENTS

SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 2

Mock Interview

COURSE NAME: Deep Learning
SLOT :A

COURSE CODE:MLA0403

COURSE OUTCOME COVERED

CO 5: Autoencoders, Undercomplete Autoencoder, Regularized Autoencoder, Sparse Autoencoder, Denoising Autoencoder, Stochastic Encoders and Decoders, Contractive Autoencoder, Deep Belief Networks (DBN), Boltzmann Machines (BM), Deep Boltzmann Machines (DBM), Generative Adversarial Networks (GANs), Generator, Discriminator, GAN Applications.

Course Outcome Weightage: 35%
Assessment Tool Weightage: 20%

Duration: 90 minutes
Mark : 25

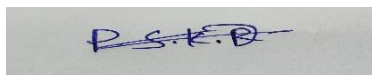
Code of Conduct:

I P Surendra Kumar Reddy (192425224) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Name of the student:P Surendra Kumar Reddy



Criteria	Subject Knowledge and Conceptual Understanding (3)	Technical Accuracy and Application of Concepts (2.5)	Problem-Solving and Analytical Thinking (2)	Communication Skills and Clarity of Explanation (1.5)	Confidence, Presentation and Professional Behaviour (1)	Total (10)
Weightage	30 %	25%	20%	15 %	10 %	100 %
Q1						
Q2						
Q3						
Q4						
Q5						
Q6						
Q7						
Q8						

Q9						
Q10						
Q11						
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Q16						
Q17						
Q18						
Q19						
Q20						
Q21						
Q22						
Q23						
Q24						
Q25						
Total						

Signature of the Faculty:

Total Marks:

QUESTIONS: 25

Total Marks:25

1. What is an Autoencoder in Deep Learning?
2. What is the main purpose of an Autoencoder?
3. What are the major components of an Autoencoder?
4. What is the role of the Encoder in an Autoencoder?
5. What is the role of the Decoder in an Autoencoder?
6. What is meant by the Bottleneck layer in an Autoencoder?
7. What is meant by Reconstruction in an Autoencoder?
8. How does an Autoencoder learn to reconstruct its input?
9. What is the difference between an Autoencoder and a traditional Neural Network?
10. What type of loss function is commonly used to train an Autoencoder?
11. Explain the basic architecture of an Autoencoder with a suitable example.
12. Why is the dimensionality of the bottleneck layer usually smaller than the input layer?
13. What happens if the bottleneck layer has too many neurons?
14. What happens if the bottleneck layer has very few neurons?
15. What is a Denoising Autoencoder?
16. How can an Autoencoder be used for image denoising?
17. How can Autoencoders be used for dimensionality reduction?
18. How can an Autoencoder be used for anomaly detection?
19. What is the difference between a Denoising Autoencoder and a basic Autoencoder?
20. Give some real-world applications of Autoencoders.
21. What is a Deep Generative Model?
22. What is the difference between a generative model and a discriminative model?
23. What is a Variational Autoencoder (VAE)?
24. What is the difference between a traditional Autoencoder and a Variational Autoencoder (VAE)?

25. What are Generative Adversarial Networks (GANs), and what are the roles of the Generator and Discriminator?

ANSWERS

- 1. What is an Autoencoder in Deep Learning?**
An Autoencoder is a neural network that learns to encode input data into a compact representation and then reconstruct the original input.
- 2. What is the main purpose of an Autoencoder?**
Its main purpose is to learn useful data representations, mainly for dimensionality reduction, denoising, and anomaly detection.
- 3. What are the major components of an Autoencoder?**
The main components are the **Encoder, Bottleneck (latent space), and Decoder**.
- 4. What is the role of the Encoder?**
The Encoder converts the input into a smaller, meaningful latent representation.
- 5. What is the role of the Decoder?**
The Decoder reconstructs the original input from the latent representation.
- 6. What is the Bottleneck layer?**
The Bottleneck is the compressed latent layer that stores the most important features of the input.
- 7. What is Reconstruction?**
Reconstruction is the process of generating an output that is as close as possible to the original input.
- 8. How does an Autoencoder learn to reconstruct its input?**
It compares the reconstructed output with the original input using a loss function and updates its weights through backpropagation.
- 9. Difference between Autoencoder and traditional Neural Network?**
An Autoencoder is mainly trained to reproduce its input, while a traditional neural network is usually trained for tasks such as classification or prediction.
- 10. What loss function is commonly used?**
Mean Squared Error (MSE) is commonly used for continuous data, while **Binary Cross-Entropy** is often used for normalized binary-like data.
- 11. Explain the basic architecture with an example.**
An Autoencoder follows **Input → Encoder → Bottleneck → Decoder → Output**; for example, a 784-pixel image can be compressed to 64 features and reconstructed.
- 12. Why is the bottleneck usually smaller than the input?**
A smaller bottleneck forces the network to learn the most important features instead of simply copying the input.
- 13. What happens if the bottleneck has too many neurons?**
The Autoencoder may simply copy the input and fail to learn a useful compressed representation.
- 14. What happens if the bottleneck has very few neurons?**
Too much compression can cause loss of important information, resulting in poor reconstruction.
- 15. What is a Denoising Autoencoder?**
A Denoising Autoencoder is trained using noisy input but learns to reconstruct the original clean input.
- 16. How can an Autoencoder be used for image denoising?**
Noise is added to images during training, and the Autoencoder learns to remove the noise and reconstruct clean images.
- 17. How are Autoencoders used for dimensionality reduction?**
The Encoder compresses high-dimensional data into a smaller latent

representation that can be used as reduced features.

18. How can an Autoencoder be used for anomaly detection?

It learns normal data patterns; abnormal inputs usually produce high reconstruction errors and can therefore be detected.

19. Difference between Denoising and basic Autoencoder?

A basic Autoencoder reconstructs the original input, while a Denoising Autoencoder reconstructs clean data from intentionally corrupted input.

20. Real-world applications of Autoencoders?

Applications include **image denoising, dimensionality reduction, anomaly detection, data compression, and feature extraction.**

21. What is a Deep Generative Model?

A Deep Generative Model learns the underlying data distribution and can generate new samples similar to the training data.

22. Generative vs Discriminative model?

Generative models learn how data is distributed and generate samples, while discriminative models learn boundaries between classes for prediction.

23. What is a Variational Autoencoder (VAE)?

A VAE is a generative Autoencoder that learns a probability distribution in the latent space and can generate new data samples.

24. Traditional Autoencoder vs VAE?

A traditional Autoencoder learns a fixed latent representation, while a VAE learns a probability distribution, enabling controlled generation of new samples.

25. What are GANs and the roles of Generator and Discriminator?

A GAN consists of a Generator that creates fake samples and a Discriminator that distinguishes real samples from fake ones; both compete during training.

PHOTO:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Subject Knowledge and Conceptual Understanding	30%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Technical Accuracy and Application of Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Problem-Solving and Analytical Thinking	20%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Communication Skills and Clarity of Explanation	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Confidence, Presentation and Professional Behaviour	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis